An effective classification of Skin Disease using Deep Learning Techniques

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Abstract--- According to WHO, Skin Infection is very common but sometimes very serious and affects a large no population all over the world. Monkeypox, Chickenpox, and Measles are the major infectious disease that causes skin infection all over the world. It has been obverse that the cases of Monkeypox have drastically increased as an effect of Covid 19. This infection has spread easily and exponentially that cause serious health issues in many underdeveloped and developing countries. Some time it has been observed that people are not able to properly classify the type of skin infection well in time, which can be a main reason of serious health issues. So, it became important to propose an effective classification of Skin Disease. In this paper the authors have proposed an effective classification of Skin Disease using Deep Learning Techniques. This approach will help in classification of chicken pox, measles, and monkeypox through skin images. The authors have utilized Monkeypox Skin Images Dataset (MSID) dataset to apply the proposed approach. The Loss, Accuracy, Precision, Recall, AUC, and F1 Score parameters have been used to analyze the performance of proposed approaches. The best algorithms with maximum accuracy and other parameters are Xception, EfficientNetV2L, and EfficientNetV2M, and CNN, VGG16, and VGG19 are the least favored algorithms for this research.

Keywords- Deep Learning (DL), Convolutional Neural Network(CNN), Machine Learning (ML), Classification, Pandemic.

I. Introduction

The cases of skin infection are drastically increasing and spreading worldwide due to various climate change issues, pollution etc. According to the WHO, in 2018 more than 140 thousand people died because of measles, on an approx. 4200 deaths per year, and 4.2 million people suffer from chicken pox. Many of the world's population suffer immensely from diseases like Chickenpox, Monkeypox, and measles [5]. The World Health Network (WHN) announced and declared the monkeypox outbreak a pandemic costing many precious human lives [16].

Table 1: Difference between chicken pox, Measles, and Monkeypox

	Chickenpox	**Measles**	**Monkeypox**
Symptoms	Rashes [first on chest, head, and back after the whole body] and other symptoms are fever, headache, weakness, low appetite	Rashes [forehead and after then move downwards]and other major symptoms are fever, running nose, small red white spots found on the mouth, red eyes	Rashes [first on the face then spread on the whole body], fever occurred before rashes
Rashes	Chickenpox rashes start with raised reddish bumps and then bumps Convert into vesicles which are filled with fluid and also cause itchiness. Before the scabbing, these bumps may get rupture	In general, Flat reddish spots are shown in the case of Measles but in the certain case raised bumps are noted and if it is so these bumps are empty or fluid less in nature	Rashes developed in the body just after 1 -3 days of high fever. Rashes start from the head of the patient and growths downwards
Transmission	Communicable in nature, Spread through cough and sneezes by patient	Communicable in nature Basically, spread through the air	When a person comes in contact with an animal that is already infected with money pox and can be transmitted from Animal to human and human to human as well

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Prevention	Best to get vaccinated	Best to get vaccinated	Avoid direct and indirect contact with an animal that may are infected or the carrier of the virus
Duration of incubation	Approx. 10 days to 21 days	Approx. 10 days to 14 days	Generally, 7-14 days but may vary from 5-21 days
Availability of vaccine	Yes, vaccine is available	Yes, vaccine is available	No
Virus	varicella-zoster virus	measles virus	monkeypox virus

The authors of this paper intend to decrease the harmful effects of these diseases through their early-stage detection. With the means of various DL-ML algorithms, this research works deals with the detection and analysis of skin infections. The authors proposed this approach by using python and its libraries namely TensorFlow and Keras. The dataset used by the authors is divided into 4 categories Monkeypox, Chickenpox, measles, and Normal Skin images. There are 109 images of chickenpox, 95 images from Measles, 285 images belonging to Monkeypox, and 293 images from a normal person. Deep learning methods have been found effective in the automated detection of skin lesions [3] and this paper deals with and detects skin abnormalities. CNN and the variants of the algorithm like VGG16 and VGG19 have also been applied along with Xception, InceptionV3, InceptionReNetV2, ResNet152V2, and EfficientNetV2M [4] [7].

This research also aims at helping the future generation to analyze and prevent the spread of such life- the best accuracy and performance and help spread it to the wider public measles, and chickenpox has sadly taken a lot many lives and impacted people. This paper deals with detecting and analyzing the best algorithms to prevent the spread of it. The authors want to detect this disease with the best accuracy and performance and help spread it to the wider public.

II. Literature Survey

Skin infections like Monkeypox, Chickenpox, and measles have been life-threatening diseases to almost the entire world and cost many precious lives. Skin disease is one of the oldest infections that humans is suffering from it due to unhealthy lifestyles, contacts and the greed of humans.

In the last 5 years, there has been a lot of research to address skin diseases as it is crucial to detect them before the widespread community transmission. Most of these research works were analyzed with features, parameters, algorithms, models used and accuracy. The authors of this paper intend to decrease the harmful effects of these diseases through their early-stage detection using the existing research studies and algorithms.

In their recent studies Vidit Kumar et al. [6] analyzed CNN and its features for Monkey Pox detection using machine learning. The dataset used by the authors was small with an accuracy of 91.11 with the Naive Bayes classifier (VGG16) as the highest. AlexNet, GoogleNet, and VGG16 were the algorithms used and a comparative study was performed. Real-time efficient network and labeling of the supervised datasets are yet to be attained.

In their research work, Md Manjurul Ahsan et al [3] discussed the damage caused by Monkeypox and its detection using modified VGG16. This research work attained an accuracy of 97 ± 1.8% (AUC = 97.2) and 88 ± 0.8% (AUC = 0.867) for Study One and Two, respectively. The post-image explanation is provided and used on the patient's dataset collected. Google Search and Data augmentation was done for data collection, preprocessing, and collection and Local interpretable model-agnostic explanations (LIME) were used along with VGG16.

In a comparative analysis, Mohammad Arafat Hussain et al [8] in their paper analyzed how AI and DL can be used to detect Covid-19, Chicken pox, Smallpox, Measles, and skin rashes. The public dataset is analyzed and applied to semi-supervised and supervised machine learning algorithms. Disease classification power of seven state-of-the-art deep from digital skin images is compared, analyzed, and differentiated.

In the studies of Shams Nafisa Ali and others [9], an ensemble of 3-fold cross-validation is worked upon the dataset. The dataset is collected from online portals, websites, social media sites, and medical data. Further, the authors of this paper data augmented and used RestNet50 achieving an accuracy of 82.96(±4.57%). The algorithms VGG16 and the ensemble system achieved accuracy of 81.48(±6.87%) and 79.26(±1.05%), respectively. A web-based application was also developed for the detection of Monkey Pox in humans.

Ijaz Gul and other authors [10] in their research analysis discuss the pros and cons of various algorithms used in the detection of Monkey Pox. The dataset used here consists of 43 images and a simple image classification model is implemented. Other biological-based detection techniques are analyzed and studied by the authors.

The authors of this paper [11] outcome better results in the detection of chickenpox diseases. The removal and elimination of false detections using the Kullback Leibler divergence are applied along with the preliminary tests for discrimination between chickenpox and herpes zoster. Circular Hough transform and Canny edge detector are applied to the model.

Miran Lee et al [12] investigated predicting Chicken Pox using LSTM AND VARIOUS OTHER Deep Learning Algorithms. The overall result accuracy is increased keyword analysis and extracting the keywords which are related to the outbreak of chicken pox. The LSTM model gave the highest correlation coefficient of 0.97114.

Manal Alghieth's [13] work develops a system of skin disease detection using the CNN technique and a pre-trained VGG19 model. The dataset used by the author contains 4500 images that were collected from different sources to train the VGG19 model. Data augmentation techniques such as zooming, cropping, and rotating were used. After that, the Adamax optimizer, which is

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most suitable for the proposed methodology, was used to obtain high accuracy and required results. This study achieved a high accuracy of 99% compared to other similar research. It can be concluded that this system is very reliable and can be integrated into smart schools.

With the means of various DL-ML algorithms, this research works deals with the detection and analysis of skin infections including Chickenpox, Monkeypox and measles. The research studies contributed to understanding the main cause, effects and DL algorithms which can be utilized in detection of these skin diseases.

III. PROPOSED APPROACH

With the help of DL Methods, the authors can easily detect Skin infection detection.

The Approach is divided into 5 parts represented by the following:

1. Data Collection: The Skin images have been taken from Kaggle. Data is already labeled in categories Monkeypox, Chickenpox, measles, and Normal Skin images. There are 109 images of chickenpox, 95 images from Measles, 285 images belonging to Monkeypox, and 293 images from a normal person.
2. Data Preprocessing: Now the next step is Data preprocessing on the label data set, In the stage. The authors do horizontal shift, width shift height shift has been performed on our dataset



Fig 1: Sample dataset Images

3. Categorize the Dataset: the dataset has been categorized into two sets: Training Set (70%) and Validation Set (30%). Firstly, the authors train our model on a training set and then the authors perform validation on our validation set. So, our training dataset has 77 images of chickenpox, 67 images of Measles,200 images of Monkeypox and 205 images of normal people. And rest images in Validation dataset
4. Model implementation: The skin infection classification model has been proposed on different 10 models. The models are CNN, VGG16, VGG19, ResNet50V2, Xception, InceptionV3, InceptionReNetV2, ResNet152V2, EfficientNetV2L and EfficientNetV2M
5. Model Evaluation: Model evaluation is done on the following parameters Accuracy, Loss, Area Under Curve (AUC), Confusion Matrix. Precise, Recall, F1-Score

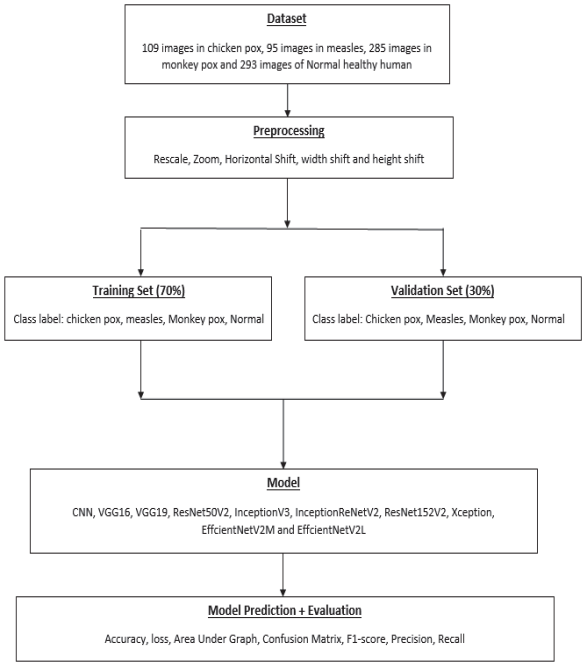


Fig 2: Proposed Approach Workflow

IV. ALGORITHMS/METHODOLOGY

With the advancement of technology, now we have a DL which has a great capability of image-based diagnoses. The authors can adapt this methodology to detect chicken pox, measles, and monkeypox. The curing method of monkeypox is start by detecting the uncommon characteristics of lesions of the skin. Monkeypox is a new disease whose symptoms include rashes on the face initially and then spread all over the body with fever. However, we can detect monkeypox using the PCR technique which stands for Polymerase chain reaction which we also used in the detection of covid19.

CNN: CNN is a deep-learning neural network that is used to solve difficult image-driven pattern recognition tasks [4].
VGG16: The Visual Geometry Group 16, is an advanced version of CNN having 16 deep neural layers. The multiple layers in the model increase depth which allows it to solve complex patterns and detect the data better. But this also increases the computational time and costs [6].
VGG19: The Visual Geometry Group 19, is an advanced version of CNN having 19 deep neural layers. It is used for object recognition and detection purposes, and since it has 19 neural network layers, it has a greater computational time [8].

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ResNet and its variants: The Residual Network helps in training hundreds and thousands of deep neural layers at a better performance. The identity shortcut connection technique that is to skip a few neural layers and is used to avoid vanishing and gradient descent problems in this algorithm.

EfficientNetV2 and its variants: are a family of image classification models, which achieve better parameter efficiency and faster training speed than prior arts. Built upon EfficientNetV1, our EfficientNetV2 models use neural architecture search (NAS) to jointly optimize the model size and training speed and are scaled up in a way for faster training and inference speed [7].

Inception-V3: The main purpose of developing this model is to deepen the Inception network. It uses 4 convolutions layers that work parallel with different sizes of the kernel with particular depth for feature extraction in images before passing into the next layer.

Xception: Xception was developed by Google in 2017. To make the convolution function more efficient in the inception block we use Xception. We achieved this by depth-wise convolution [17].

InceptionResNet: It consists of 449 layers which include convolutions layers, pooling layers, batch normalization, and so on. It is the combination of Inception architecture with the residual connection. By this, we add both model advantages into one model [17].

V. Result and Discussion

The approach which the authors proposed is undergoing an evaluation phase and its evaluation was done on the training and testing dataset. The models are also judged based on Accuracy, Losses, F1 Score, Precise Area under the graph, and Recall. The dataset on which the authors are working has been taken from Kaggle and it contains 4 classes chicken pox, Monkeypox, Measles, and Normal and the total number of images is 782. The deep learning model which the authors proposed has been implemented using 10 models CNN, VGG16, VGG19, ResNet50V2, Xception, InceptionV3, ResNet152V2, InceptionReNetV2, EfficientNetV2M, and EfficientNetV2L. The model is executed with 50 epochs and the step per epoch was 10.

Table 2: Training Result

MODEL	LOSS	ACCURACY	PRECISION	RECALL	AUC	F1_Score
CNN	119.91	51.02	89.23	10.72	74.16	28.97
VGG16	91.96	63.95	75.66	52.86	85.60	37.25
VGG19	95.9	62.84	72.162	49.35	83.99	36.17
RESTNET50V2	6.784	97.60	97.95	97.23	99.92	96.76
InceptionV3	2.69	99.63	99.63	99.63	99.86	99.41
ResNet152V2	30.98	88.35	91.55	86.13	98.39	83.39
InceptionResNetV2	1.56	99.63	99.63	99.63	99.91	99.51
Xception	1.76	99.45	99.45	99.45	99.99	99.01
EfficientNetV2L	1.69	99.63	99.63	99.63	99.99	99.51
EfficientNetV2M	2.72	99.08	99.07	98.71	99.99	98.90

Table 3: Testing Result

MODEL	LOSS	ACCURACY	PRECISION	RECALL	AUC	F1_Score
CNN	117.75	56.85	90.63	12.03	77.15	32.49
VGG16	96.02	64.35	71.19	52.28	83.85	37.45
VGG19	104.39	59.33	68.55	51.45	80.46	34.17
RESTNET50V2	74.23	81.75	83.33	80.91	94.103	76.83
InceptionV3	56.97	89.67	89.58	89.21	96.65	87.27
ResNet152V2	61.87	80.08	83.33	76.76	94.03	71.41
InceptionResNetV2	91.93	87.14	87.14	87.14	94.94	84.20
Xception	57.03	89.21	89.87	88.38	97.120	85.83
EfficientNetV2L	33.67	92.53	92.50	92.12	97.69	90.93
EfficientNetV2M	47.62	86.72	88.085	85.89	96.58	83.13

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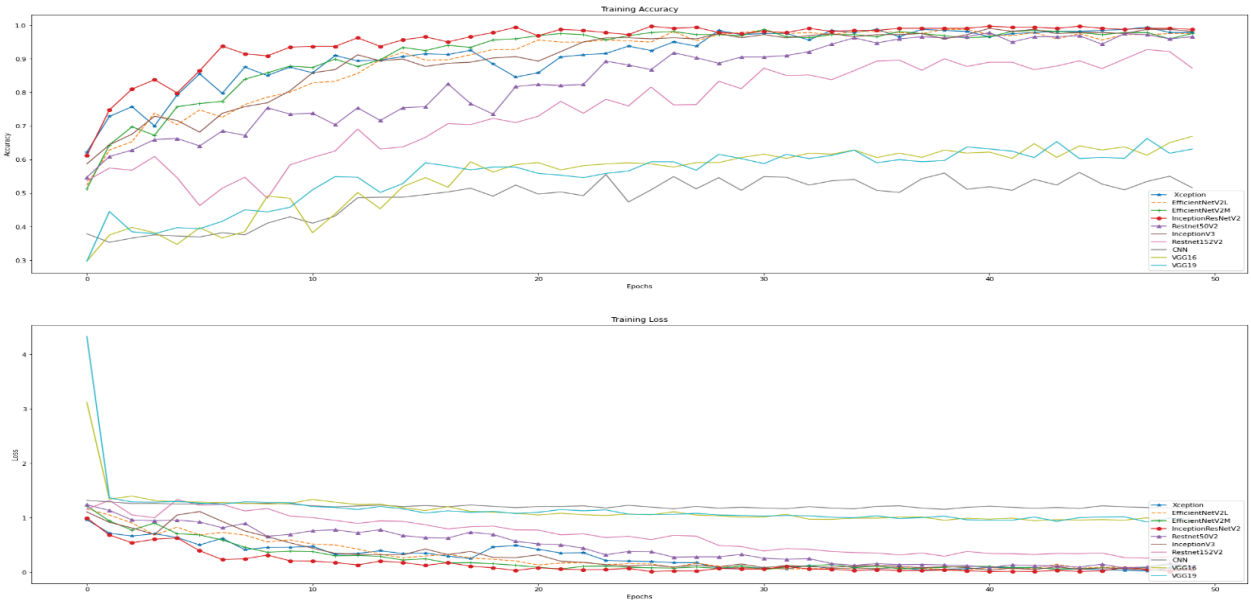


Fig 3: Graph

The performances of these models are compared among themselves. The Basic CNN-based model got 56.846 % accuracy, VGG16 based model prediction was 64.315 & accurate while VGG19 based model got an accuracy of 59.336 %. The ResNet50V2 got an accuracy of 81.742%. The model based on ResNet152V2, InceptionV3, and InceptionReNetV2 got accuracy of 80.082%, 89.626%, and 87.136% respectively. The model Based on Xception, EfficientNetV2M, and EfficientNetV2L has achieved an accuracy of 89.211%, 86.721%, and 92.531%.

VI. Conclusion

In this paper, the proposed system compares CNN, VGG16, VGG19, InceptionV3, InceptionReNetV2, ResNet152V2, EfficientNetV2L, EfficientNetV2M and Restnet50-based models for the classification of Skin infection. From the above result, the authors can say EfficientNetV2L has performed better in terms of accuracy. The authors of this paper did a comparative analysis of the detection of skin abnormalities and detection of skin diseases.

In near future, more images from this category will be collected and make a larger dataset as from now and the given approach might be used for the correct detection of skin infection. The authors also aim at increasing the overall performance and accuracy of the model in the future and using a wider dataset.

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Imputation Techniques Analysis for Incomplete Medical Datasets in Case-Based Reasoning System

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***Abstract*— Case based reasoning (CBR) based expert systems are widely accepted for healthcare-related applications. CBR is analogous to how humans solve problems using prior experience. CBR is a lazy learner and maintains the cases in case-base to solve the encountered target case. The accuracy of any case based reasoning based system depends upon the quality of cases in its case base. Nowadays, an ample amount of electronic healthcare data is available, but unfortunately, electronic data come along with a growing amount of missing data. Multiple techniques are available to handle those missing values. One widely used way is to use imputation methods to compute missing values for different attributes, which utilize the total actual values in the given dataset. The study was conducted to find the best suited imputation technique for calculating missing values for constructing case base in case-based reasoning system. This study uses benchmark medical dataset to examine four combination imputation techniques and data pre-processing method. The author proposes a hybrid model that performs the pearson correlation to find the critical feature and then used imputation techniques to fill in missing values in different attributes. Experiment results show that the Pearson correlation and linear regression (LR) outperform the other 3 imputed techniques' and give more accurate imputed values. To justify the implication of data pre-processing methods with imputation technique, the findings are carried to medical dataset collected from UCI.**

Keywords: Case based reasoning System (CBR); imputation technique; quality of case-base; regression technique; preprocessing; medical dataset; case-based maintenance; missing data; correlation-based feature selector

I. INTRODUCTION

Case Based Reasoning (CBR) is a methodology that uses knowledge obtained from past incidents (also called cases) to solve a given target problem. The CBR cycle [22], as shown in Figure 1, is split into four main phases known as 4R; the first phase is to retrieve the most related case, the second phase is to reuse the knowledge of the retrieve case to solve the encountered problem, the third phase is to perform the revision of the proposed solution if required and 4th phase is to retain the solved solution to handle similar cases in the future. Several works[7] proposed suggestions for learners through case-based reasoning systems; however, these CBR works are not aimed at data-pre-processing algorithms for constructing and maintaining a case base. The performance of any system using Case Based Reasoning (CBR) is based on how correctly related cases are retrieved from the case base in the retrieval phase [23]. The performance of the retrieval step affects all the subsequent phases. Recent developments in Information Technologies e.g. Internet of things, social networks, mobile applications etc. generate a massive amount of electronic healthcare records (EHR). Regrettably, expanding databases come with an increasing volume of missing data that reduces the accuracy of CBR based system since incomplete cases cannot be considered as reliable as complete ones. In case-base, missing data can reduce CBR's ability to create accurate predictions. EHR data cause considerable challenges because of not been unambiguously accumulated for research objectives and is subject to a significant amount of missing values. Improper handling of missing values can lead to considerable biases in results. These issues will be more visible as EHRs become more prevalent, as the amount of discrete electronic data increases and access to this data becomes increasingly available. Additionally, cases collected from different hospitals might have different attributes. To combine the dataset having different attributes required to impute values for missing attributes. In the context of CBR, very little work has been done to thoroughly analyze the effect of missing data and solutions to confront this issue. In particular, a generalized solution that can process data under different missing conditions for different variable types is missing. This paper proposes a combined approach of pre-processing with an imputation technique for case-based reasoning (CBR) framework for an incomplete case base that provides correct similar cases. Our study shows that missing data must be handled when a correct CBR outcome is required.

The remaining paper is organized as follows. Section 2 discusses the related work. Section 3 proposes the framework to handle missing data in CBR. Section 4 explains the experimental setup and

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results are presented in section 5 and conclusion is drawn in section 6.

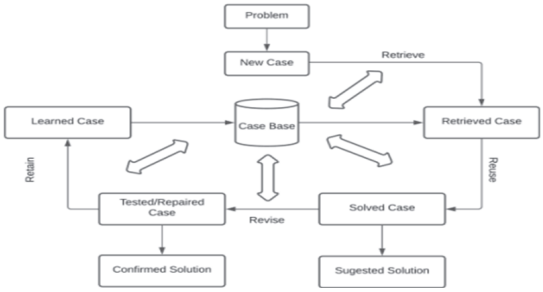


Fig 1. CBR Cycle

II. Background Study

This section highlights and discusses the data missingness in CBR, the types of missing data, and the theoretical concepts required to understand how to handle missing data.

A. Understanding of data missingness in a Case Base Reasoning System

Cases stored in case bases in CBR-based systems can be broadly categorized into structured and non-structured cases. Structured cases primarily consist of measurable numeric values (e.g., temperature, blood pressure) and distinct values, the integers or characters for predefined classifications (e.g., different codes used for diagnosis based on ICD-9). Non-structured /text cases, which is collected from the textual part of the patient chart (e.g., progress note given by physician). Whether cases are structured or unstructured, missing data is of significant concern and can appear in different ways. Missing values for different attributes can occur due to a lack of proper collection of data (e.g., while collecting data, the patient is asthmatic or not is not asked and reported) or issue in record maintenance (e.g., the patient is questioned for asthma but missed recording the response in the medical record). Lack of proper documentation is the most common concern for a patient not having an indication. In place of storing a negative value for each possible indicator, all attribute fields having negative values are not filled, resulting in missing values. The current research focused on structured cases. The required situation is to enhance the completeness and accuracy of structured cases in the case base [1]. To address the stated problem, this research analysed imputation technique which is combined with feature selection for incomplete medical dataset in case-based reasoning (CBR) system.

B. Types of missing data in the medical dataset

Usually, missing values in medical cases happen for different reasons and are divided into 3 categories. M**issing completely at random (MCAR)** means that data missing has nothing to do with the person from whom data is collected. For example, a questionnaire that is used to collect data might get lost in the post, or a blood sample might get harmed in the lab. In CADET, information of sex might be MCAR. M**issing at random (MAR)** means that the missing values are related to the person but can be depicted from other information related to the person. It is not explicitly related to the missing values. For example, if a child missed his academic assessment because of his illness, this information might be deducted from other data collected about the child's health, but it is not related whether the information about the child's illness is collected or not. When data are **missing, not at random**, the missing value is associated explicitly with what is actually missing, e.g., if a patient does not join a drug test because that patient took drugs the night before. The problem is deciding which of these situations we have; in the same dataset, we may have missing data for each reason [24]. The current research focused on MCAR.

C. Treating Datasets having Missing Data

Missing data is inevitable. Various imputation techniques have been used to handle missing data, which could be broadly classified as single imputation or multiple imputations. Single mutation generates a single value for the missing value and has less computation cost. Some of the widely used single imputation techniques are as follows [2,3].

(1) Listwise deletion: Listwise deletion deletes the entire case that contains a missing value. Listwise deletions results in reduced sample size.
(2) Zero imputation: It assigns some incorrect values to the missing value. Like in this case zero is assigned to all the missing values.
(3) Mean imputation assigns the average value of the attribute to the missing value.
(4) Multiple imputation uses the values from all the attributes to find out the value for missing data.
(5) Linear Regression: The linear function is measured using the present values of the same attributes. Linear function is used to calculate missing value.
(6) Hot deck: Missing values is replaced with the most similar case
(7) KNN: missing value is replaced with the feature value of most similar K instances.
(8) K-means clustering: K- clusters are formulated from non-missing data, and then the missing value is imputed using these clusters.

Using different simulation models, multiple imputations generate multiple values for the same missing value. Each missing value is replaced with n values obtained from n iterations (where n >1 and n

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usually lies between 3 and 10). A few multiple mutation techniques are discussed as follows:

(1) Multivariate Imputation by Chained Equation (MICE) [5, 6, 8, 9, 10 and 11]: It provides multiple values in the place of one missing value by generating a series of regression or other suitable models. Missing data is treated as dependent variable and other variables are known as independent variable.

(2) Amelia [4]: It uses a bootstrapping and expectation-maximization algorithm to impute missing value. It creates multiple imputations by multiple iterations [4].

D. Literature Survey

This section discusses the related work. Table 1 highlights the research carried out by different authors related to the proposed work, the technique implemented by different authors, a short description of the work, the dataset used for the experiment work and their conclusion for the current paper. The extensive literature survey shows that little research is conducted in this area.

TABLE 1 BRIEF DESCRIPTION OF PREVIOUS RESEARCH

Research Paper	Algorithm used	Brief description of the research	Data set	Implication on the current paper
R. Jagannathan and S. Petrovic (2009), [12]	Filter imputation algorithm is used	The author identified the feature ln with the highest correlation with the missing feature value lm. Then, the author proposes to consider all the cases in the case base with the same value for attribute ln, use the mode value of attribute lm from the extracted case, and use that value as the imputation value.	Prostate Cancer	It offers an excellent alternative to other widely used techniques to delete cases with a missing value.
Nikolas Löw (2019), [13]	multiple retrieval case-based reasoning framework	The author proposes a framework that uses K-imputed datasets. Each dataset has different imputed values generated from different algorithms. The research concluded that the mean method performs best, followed by imputation by random forest and then followed by CART.	Liver	It has been observed that missing data should not be ignored to extract proper cases to handle the target problem.
Chlioui, I., et. al (2021). [14]	CBR ,DT, SVM ,KNN, RF, multilayer perceptron	The author evaluates the accuracy of 5 different ensembles CBR to handle nine different missing % rates. The study proved that RF-CBR outperforms the other 5 ensemble CBR for algorithms for breast cancer detection.	Breast Cancer	This research proved that classification accuracy was higher when using ensemble CBR to handle missing data than the CBR alone.
Souad Guessoum, et. al [15]	Online optimistic, pessimistic and medium ,offline statistical and online CBR approach	The author uses the online strategy to estimate the cost for the local similarity of a feature whose value is missing. The study concluded that a pessimistic online strategy generates the best results that estimate the similarity by the possible smallest value. For offline strategy, the missing value of a feature is estimated by the mean values of other similar cases, and their similarity measure is used to allocate weight.	Chronic Obstructive Pulmonary Disease	The critical issue in the case-base that many cases contains missing values
Pedersen, A., et. al (2017). [16]	Multiple Imputation	The author highlights the type of missing data and compares the traditional methods and multiple imputations to handle missing data.	Epidemiological Research	Multiple imputation gives the more unbiased estimates.
Wang, H. C., et. al (2012). [17]	Association rules	The author proposed association rules to compute the missing values of different attributes, and the rule confidence is referred to validate the attrribute weight .	UCI Repository	Classification can be performed before mining rules to enhance the quality of the rules.
Abnane, I., et. al (2020). [18]	Fuzzy based	The autgor proposed a fuzzy-based cbr system and compared the propsed system with the KNN imputation technique.	Software engineering dataset	Missingness mechanism has impact on both the techniques.
Dzulkalnine, et. al. (2019). [19]	FPCA along with SVM Fuzzy C means	Author proposed a hybrid model to minimze ccomputatio. FPCA used for feature selection and fuzzy c means as too impute missing data.	Diabetes dataset	By removing irrelevant features imputation performance can be improved.
Rani, P., et. al (2021). [20]	KNN, multivariate imputation, mean and mode	The author compared the 4 imputation techniques , trandom forest (rf), the Support vector machine, Logistic regression and Naïve Bayes. The author concluded that multivariate imputation by chained equation performed best.	Heart dataset	Root mean square errors were compared to find the best combination
Xia, J., et.al (2017). [21]	Adjusted Weight Voting for Random Forest	Author proposed the model which Adjusts the Weight for Voting Random Forest technique that performs classification without performing imputation of missing values	Liver, Diabetes, breast cancer, Heart	More efficient. performed classification without performing imputation

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III. PROPOSED WORK

Missing data impacts the performance of a case-based reasoning system in terms of accuracy. To enhance the performance of a case-based reasoning system, hybrid model is proposed to handle missing data in the cases stored in case base as shown in Figure 2. It has been observed that the irrelevant features increases the computation cost for processing data. Therefore, the proposed model removes the irrelevant features first using the feature selection method and then imputes the missing data to store in the case base.

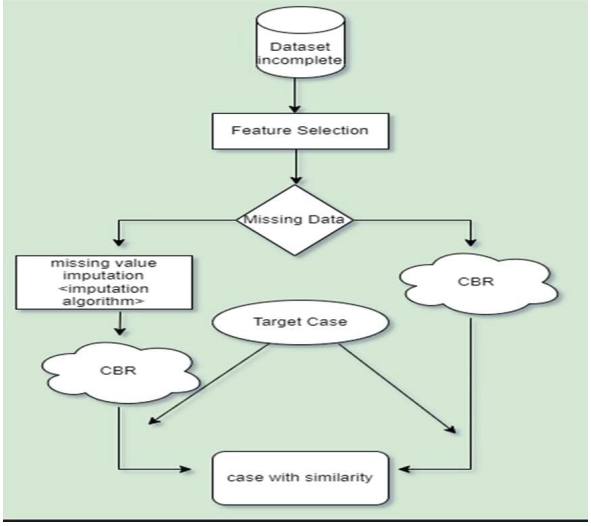


fig 2. Proposed framework for handling incomplete dataset for CBR

IV. EXPERIMENTAL SETUP

This section describes the dataset, implementation of the proposed work, and the performance measures to evaluate the different techniques.

A. Dataset

For experimentation, liver patient data is collected from UCI data repository. Dataset contains 583 cases where each case has 10 attributes namely age, SGOT, total Bilirubin, gender, direct Bilirubin, albumin, total proteins, SGPT, A/G ratio and Alkphos. Of 583 cases, 416 cases belong to liver patient, and 167 are for non-liver patient cases.

B. Implementation

The liver dataset used for experiment work is normalized using the Z -Score method. The z -Score method normalizes the attribute values in the range of -1 to 1 using equation (1) where a is the attribute value, m is mean and sd is standard deviation.

$$z - score = \frac{a-m}{sd} \qquad (1)$$

After normalization, the proposed model uses the feature selection method to select important features. This study uses the Pearson correlation method to identify highly correlated features. For example, from the pearson correlation method, it has been observed that direct bilirubin and total bilirubin is highly correlated with a value of +.875. So, between these 2 features, direct bilirubin feature is removed from the dataset. After that the dataset is divided into 2 parts D_Train for training and D_Test for testing the imputed models in the ratio of 70 and 30 % . D_Train dataset does not contain any missing values with the complete dataset, whereas in 10 % of total D_Test instances missing values are introduced. The focus of the study is on the MCAR type of missing value. The four models: Support Vector Machine, Random forest, Neural Network, and Linear regression, are used and compared to impute missing values in different attributes. The hyper-parameters used for different models are stated in Table 2. The 10-fold cross-validation method is used to reduce the biases of the model. The results of different models are presented in Table 3.

TABLE 2: HYPERPARAMETETR USED FOR DIFFERENT MODELS

S.NO	ALGORITHM NAME	PARAMETER
1	Support Vector Machine	Sigmoid Kernel
2	Random forest	Max. depth of tree is 5
3	Neural Network	Hidden Layers are 5 and at max. iteration is 1000

C. Performance Measure

The performance of different imputation techniques is compared using different statistical measures, namely mean absolute error (MAE), root mean squared error (RMSE), mean squared error (MSE), R square (R^2)[25].

1) Mean Squared Error

It is a statistical measure that computes the amount of error that evaluates the average squared difference between the imputed and actual values. MSE can be calculated using the equation as shown in equation (2).

$$MSE = \sum_{i=0}^{n} \frac{truevalue_i - imputed_value_i}{n} \qquad (2)$$

2) Root Mean Squared Error (RMSE)

RMSE is a statistical measure that computes the standard deviation of the difference between the observed and imputed observations and is calculated using the equation given in equation (3).

$$RMSE = \sqrt{\sum_{i=0}^{n} \frac{truevalue_i - imputed_value_i}{n}} \qquad (3)$$

3) Mean Absolute Error(MAE)

It is a measure that computes the average variation between the observed and imputed value and is calculated using equation (4).

$$MAE = \left| \sum_{i=0}^{n} \frac{truevalue_i - imputed_value_i}{n} \right| \qquad (4)$$

4) R Square

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It specifies how well the data fit the regression model (goodness of fit) and is calculated using Equation (5).

$$R^2 = \frac{Sum\ of\ squares\ due\ to\ regression}{total\ sum\ of\ squares} \quad (5)$$

V. Results

This section compares the performance of different imputation techniques for the proposed hybrid model using the Indian liver patient dataset. The proposed framework uses a combination of feature selection and imputation techniques to handle missing attribute values and improve the performance of a case-based reasoning system designed to diagnose liver disease. Firstly, the feature selector method is used to find the essential features. In this study, Pearson correlation is used to find the correlated attributes. Correlated attributes are removed from the dataset. Furthermore, missing values are imputed using SVM, linear regression, random forest, and random forest. To perform the simulation 10% values are removed from each attribute and then imputed using different imputation technique. The simulation results are evaluated using performance measures known as RMSE, MSE, MAE, R2 as shown in Table 3. MSE, RMSE and MAE values indicate how closely the imputed values are to actual value. Model having less value for all the stated performance measures are considered as best model. From the simulation results it has been observed that combination of pearson correlation and linear regression outperforms the SVM, Neural Network and Random forest imputation technique with lowest value for MSE i.e. 0.023, 0.151 for RMSE and 0.1 for MAE. R – Squared statistical measure indicates how well regression model fits the data. 0.99 indicates the good R-Squared value for linear regression model.

TABLE 3: PERFORMANCE MEASURES FOR IMPUTING MISSING VALUES USING 4 DIFFERENT MODELS

Model	MSE	RMSE	MAE	R2
SVM	1.448	1.204	0.611	0.338
Random Forest	0.396	0.629	0.251	0.819
Neural Network	0.404	0.636	0.382	0.815
Linear Regression	0.023	0.151	0.1	0.99

Figure 3. shows the graphical representation of various performance indicators of different models. MSE, RMSE, MAE has smallest and highest value for R2 for linear regression model.

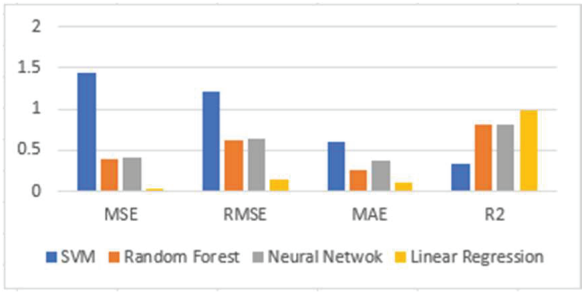


Fig 3. Graphical representation of performance indicators

VI. Conclusions

The accuracy of any case-based reasoning system depends upon the quality of retrieved cases. Different cases might have different attributes as they are collected from different hospitals. Furthermore, few attributes might have missing value due to versatile reasons. To enhance the quality of retrieved cases in CBR, one must handle attribute missing values. This study proposes a framework that combines feature selection and imputation techniques to handle incomplete cases. From the experimentation, it has been concluded that the combination of Pearson correlation and linear regression outperforms the other combination of correlation and imputation techniques, namely SVM, Random Forest, and Neural Network.

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